

CoRoT-1b

R-2569

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-1_180207 · created 2026-08-26T23:48:10+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2458156.7062 +/- 0.0038 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.143 +/- 0.041
Transit depth (Rp/Rs)^2	2.03 +/- 1.17 [%]
Orbital Inclination (inc)	88.85 +/- 2.84
Airmass coefficient 1 (a1)	1.9 +/- 0.6
Airmass coefficient 2 (a2)	-0.53 +/- 0.45
Scatter in the residuals of the lightcurve fit is	30.4 %
Best Comparison Star	None
Optimal Aperture	3.0
Optimal Annulus	7.5
Transit Duration (day)	0.112 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.143 ± 0.041 vs 0.1388 (deviation 0.09σ)
Duration fit vs published	0.112 d vs 0.1046 d (deviation 1.14σ)
Mid-time O–C	-6.8 min
Depth SNR	3.2
Residual scatter	30.4 %

Light curve

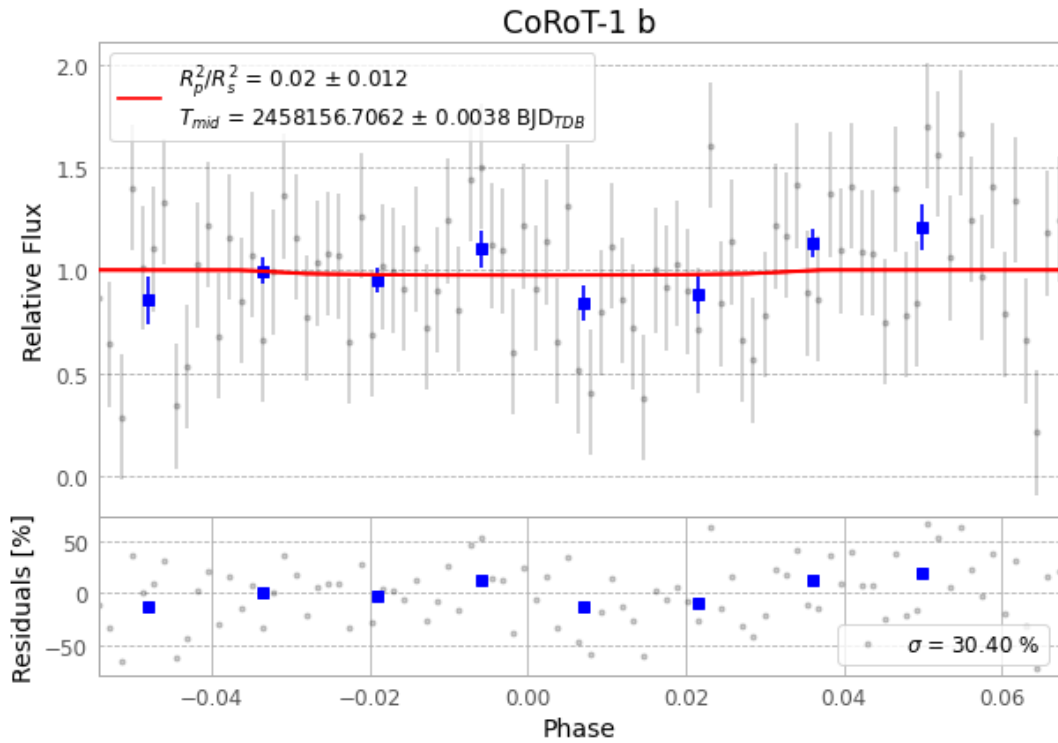


Figure 1 — FinalLightCurve_CoRoT-1 b_2018-02-06.png (SHA-256 9e6993e070ff5757...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [283, 103], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 99875c5c68f9e940...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

89 FITS frames (58 MB), CoRoT-1180207025112.FITS → CoRoT-1180207071512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-1-180207.log (427a0135c319...), FinalLightCurve_CoRoT-1 b_2018-02-06.png (9e6993e070ff...), FinalLightCurve_CoRoT-1 b_2018-02-06.csv (d2bfb930bf2c...)

Compute resources

Wall time 92.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-26 23:48Z.